

Lab_Day06-Pentesting Basics

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Task 1: Using VPN Connection

Use THM OpenVPN room to establish VPN connection between your Kali Linux attack box and TryHackMe network to access target machines. This room will guide you about connection process. Use relevant connection guide to connect your machine to THM machines for future exercises.

From you Kali Linux, visit: <https://tryhackme.com/room/openvpn> and complete to room (CAPTURE YOUR FLAG).



✓ Visit <https://tryhackme.com/access> which should show you **Connected**.

Task 2: VPN Connection Best Practices

Understand key points about how to manage your VPN connection and troubleshoot common issues.

1. To Connect: `sudo openvpn yourusername.ovpn` in terminal
 - To Disconnect: Press **Ctrl + C** in the OpenVPN terminal window
 - Alternative: Close the OpenVPN terminal window
 - VPN does NOT auto-connect on reboot. You must manually reconnect each time you start Kali Linux.
 - Keep the .ovpn file in an easily accessible location
 - Multiple terminals workflow:
 - Terminal 1: Keep OpenVPN running (don't close)
 - Terminal 2: Use for commands and tools (nmap, gobuster, etc.)
 - Terminal 3+: Open additional terminals as needed for multitasking

- ✓ You understand how to connect and disconnect from VPN
- ✓ You know how to verify active connection

Task 3: Hack the Target

- Make sure you are connected to THM via openvpn: <https://tryhackme.com/access> -> Note your tun0 IP (e.g. 192.168.150.194)
- Run target machine <https://tryhackme.com/room/blue>
- Note its IP: 10.80.151.188 (it will be different for you)

don't forget to Add 1 hour if task not completed in 1 hour)

AttackBox: Kali Linux running on your Laptop (inside VirtualBox or VMware)

Target: TryHackMe Machine

Task 4: Scan the Target

```
$ nmap -sV -vv --script vuln 10.128.177.69
```

- **-sV (Service Version Detection):** probes open ports to determine what software is running and its specific version number.
- **-vv (Double Verbosity):** tells Nmap to be "very chatty." Instead of waiting until the end to show you results, it will give you real-time updates on what it's finding, which scans are completing, and any errors it encounters.
- **--script vuln:** This tells Nmap to run a specific category of scripts from the **Nmap Scripting Engine (NSE)**. The vuln category checks the target for known, high-impact vulnerabilities (like Heartbleed, EternalBlue, or outdated SSL versions).

- To save your Nmap scan results to a file and later import them into Metasploit, you want to use Nmap's XML output format. Metasploit parses XML perfectly, allowing it to reconstruct the entire host, port, and service architecture in its database. Here is the step-by-step process:

Step 1: Save the Nmap Output using -oX

The flag to save Nmap output as an XML file is -oX (lowercase o, uppercase X).

Modify your original command to include the flag and a filename like this:

```
nmap -sV -vv --script vuln -oX my_scan_results.xml 10.128.177.69
```

(Alternatively, you can use -oA filename which saves the results in three formats simultaneously: normal text .nmap, grepable .gnmap, and XML .xml).

Step 2: Load the XML file into Metasploit

Once your scan finishes and you have your my_scan_results.xml file, open Metasploit:

```
msfconsole
```

Inside msfconsole, use the db_import command followed by the path to your XML file:

```
msf > db_import my_scan_results.xml
```

Metasploit will parse the file, and you will see an output saying something like: [*] Importing 'Nmap XML' data... [*] Successfully imported /path/to/my_scan_results.xml

Step 3: Verify the Import

To make sure everything loaded into your database correctly, run your standard database viewing commands:

```
msf > hosts
```

```
msf > services
```

Your targets and all their discovered vulnerabilities/services will now be populated and ready for you to attack!

Task 5: Find and Research Vulnerabilities

- Search in ExploitDB, Github (ms17-010)
- A command line tool in Kali Linux for ExploitDB is searchsploit. Try this command as well:
\$ searchsploit ms17-010

Task 6: Initialize database to be used with MSF

We can perform nmap scanning inside msf which will be useful here because we will have the findings of the scan in the framework which we can exploit using its modules.

1. Start the PostgreSQL database service

```
systemctl start postgresql
```

2. Enable it so it starts automatically every time you boot Kali

```
systemctl enable postgresql
```

3. Initialize the Metasploit database framework

```
msfdb init
```

Task 7: Exploit vulnerability

Use msfconsole to exploit vulnerability and gain access to the machine.